CS & IT ENGINEERING

Operating System

File System

Lecture - 3

By- Vishvadeep Gothi sir

Recap of Previous Lecture

Topic Disk Blocks

Topic File Allocation Method

Topic Disk Scheduling

Topics to be Covered

Topic Disk Scheduling

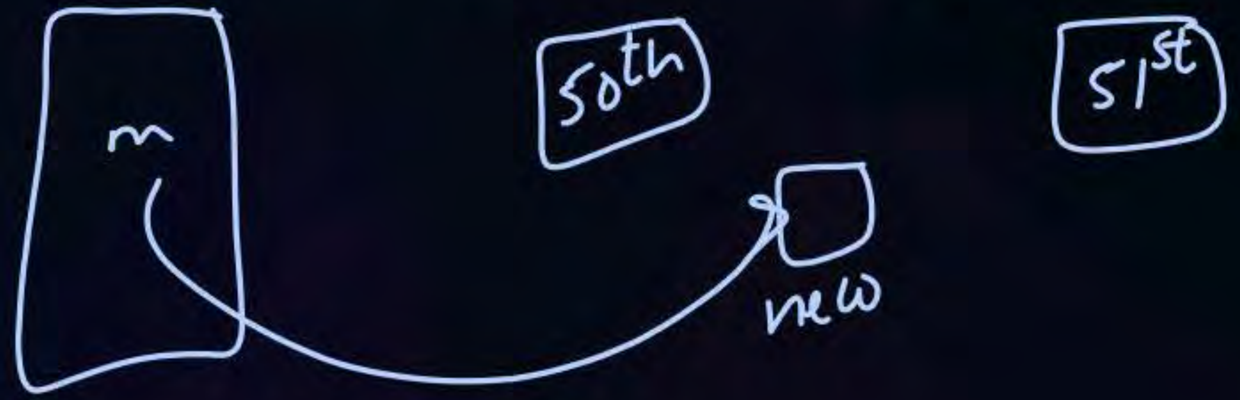
Topic FCFS, SSTF

Topic Scan, Look Algorithms

Topic : Question

[GATE-2022]

#Q. Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50^{th} and 51^{st} block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B respectively, then the value of $n_A + n_B$ is 153 ?



$$\begin{array}{r} 101 \\ 52 \\ \hline 153 \end{array}$$

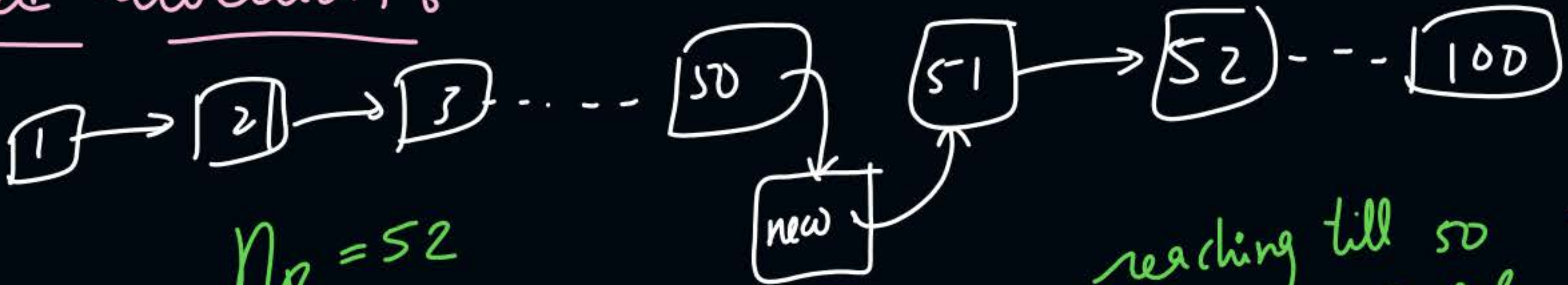
A) **Contiguous allocation :-** $n_A = 101$

shift block 51 to 100, one position right $= 2 * 50 = 100$

new block content copy $= 1$

$\overline{101}$

B) **Linked allocation :-**



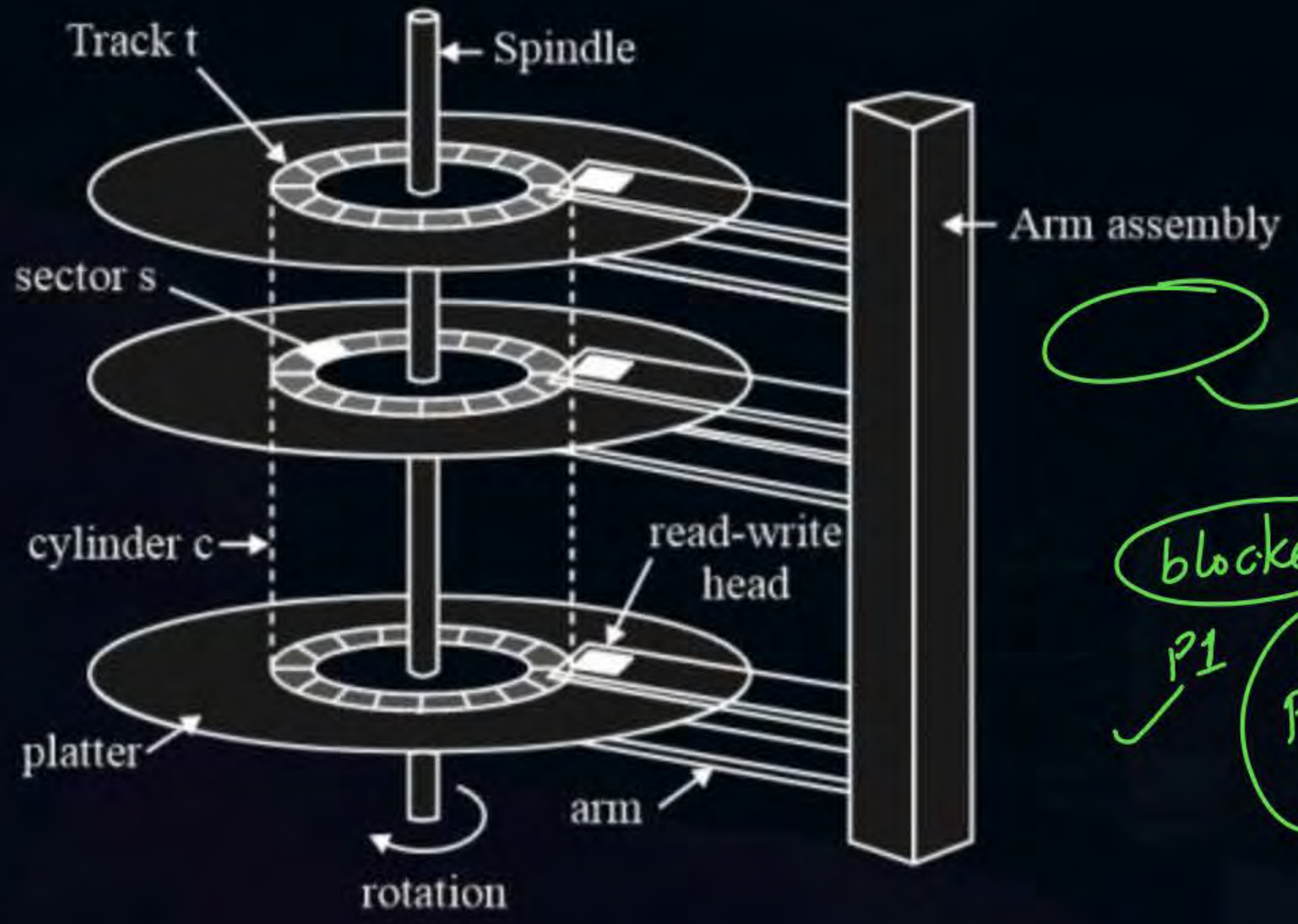
$n_B = 52$

reaching till 50 $\Rightarrow 50$

updating content & links $\Rightarrow 2$

$\overline{52}$

Topic : Disk



Running

blocked

P1

P2
P3

Topic : Cylinder

- Collection of tracks of same radius from all surfaces

Topic : Disk Scheduling

- Done by operating systems to schedule I/O requests arriving for the disk

⇓

to service the pending disk requests in such a way that the overall performance of the system can improve.

Topic : Disk Scheduling Algorithms

1. FCFS (First Come First Serve)
2. SSTF (Shortest Seek Time First)
3. Scan
4. C-Scan (Circular-Scan)
5. Look
6. C-Look (Circular-Look)

Topic : FCFS (First Come First Serve)

Assume 200 cylinders in disk numbered 1 to 200

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

The Read/Write arm is at 50

service the requests in their order of arrival.

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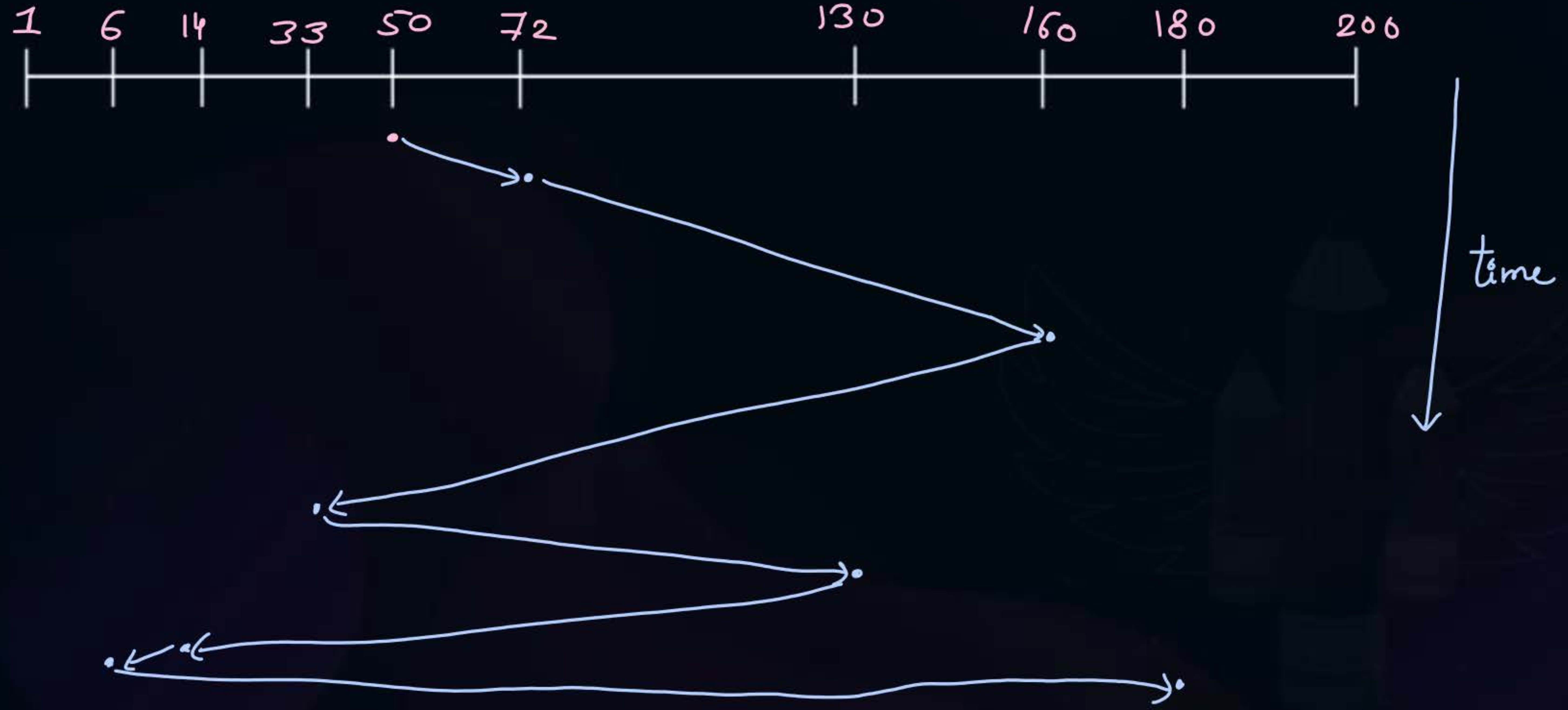
no. of disk head movements $= (72-50) + (160-72) + (160-33) + (130-33)$

$+ (130-14) + (14-6) + (180-6)$

$= 632$

Topic : FCFS (First Come First Serve)

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



Ques 1)

if moving head for 1 cylinder takes 0.5 msec on an avg.,

then seek time of disk = $632 * 0.5$ ms

= 316 msec

Ques 2)

1 head movement = 0.1 msec

1 time direction change takes ⇒ 5 msec

total seek time = $(632 * 0.1 \text{ msec}) + (4 * 5 \text{ msec})$

= 83.2 ms

Topic : FCFS (First Come First Serve)

Advantages:

- Every request gets a fair chance
- No indefinite postponement (no starvation of any request)

Disadvantages:

- Does not try to optimize seek time
- May not provide the best possible service

Topic : SSTF (Shortest Seek Time First)

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

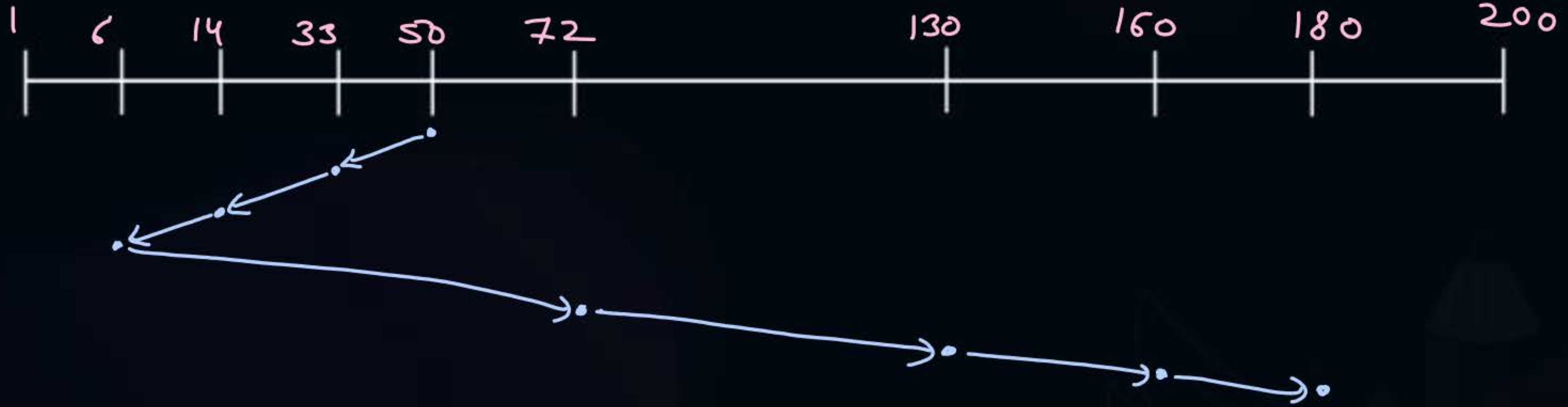
The Read/Write arm is at 50

service the request of cylinder, which is nearest from current position

tie breaker ⇒ given in question

Topic : SSTF (Shortest Seek Time First)

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



no. of head movements $= (50-6)+(180-6)$

$= 218$

Topic : SSTF (Shortest Seek Time First)

Advantages:

- Average Response Time decreases
- Throughput increases

Disadvantages:

- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has higher seek time as compared to incoming requests
- High variance of response time as SSTF favors only some requests

Topic : Scan (Elevator)

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

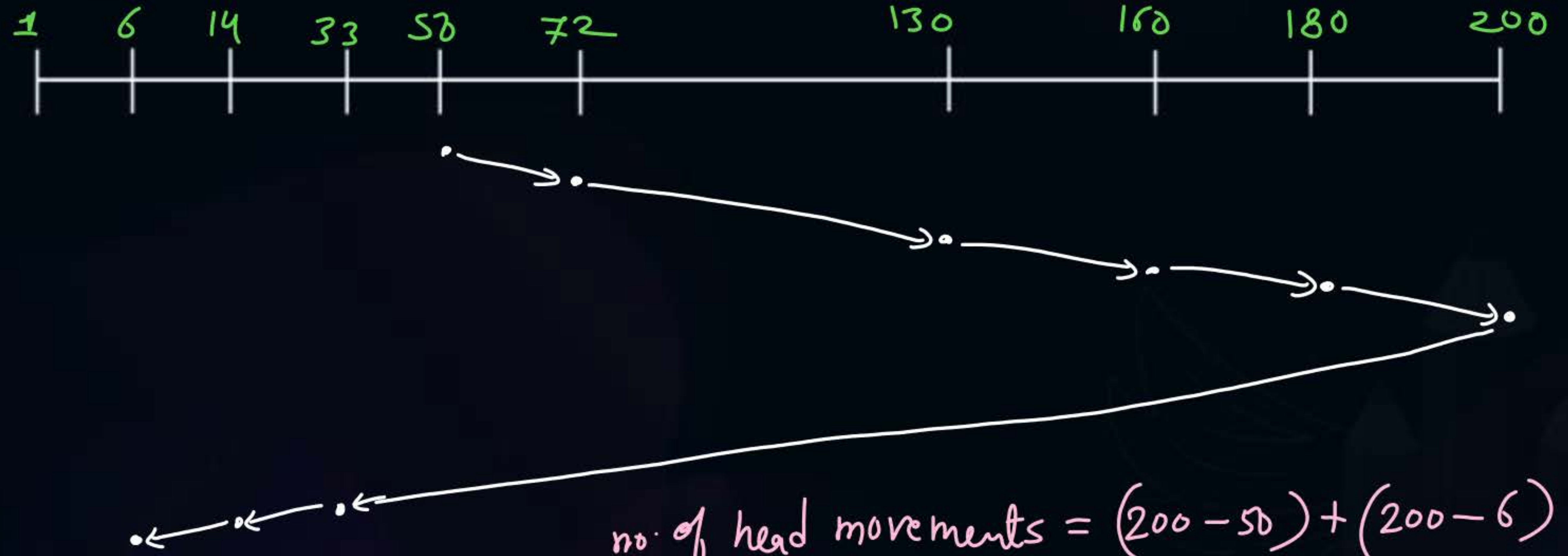
The Read/Write arm is at 50,

The arm should move "towards the larger value"

move towards one direction and service all pending requests on the way. Change direction at last cylinder and move into other direction to service requests on the way.

Topic : Scan (Elevator)

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



no. of head movements $= (200-50) + (200-6)$

$= 150 + 194$

$= 346$

Topic : Scan

Advantages:

- High throughput
- Low variance of response time
- Average response time is high

Disadvantages:

- Long waiting time for requests for locations just visited by disk arm

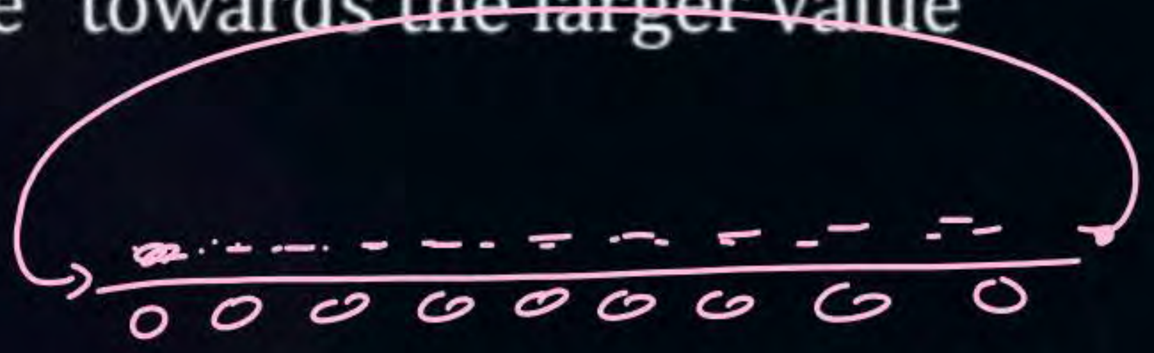
Topic : C-Scan

circular scan

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

The Read/Write arm is at 50,

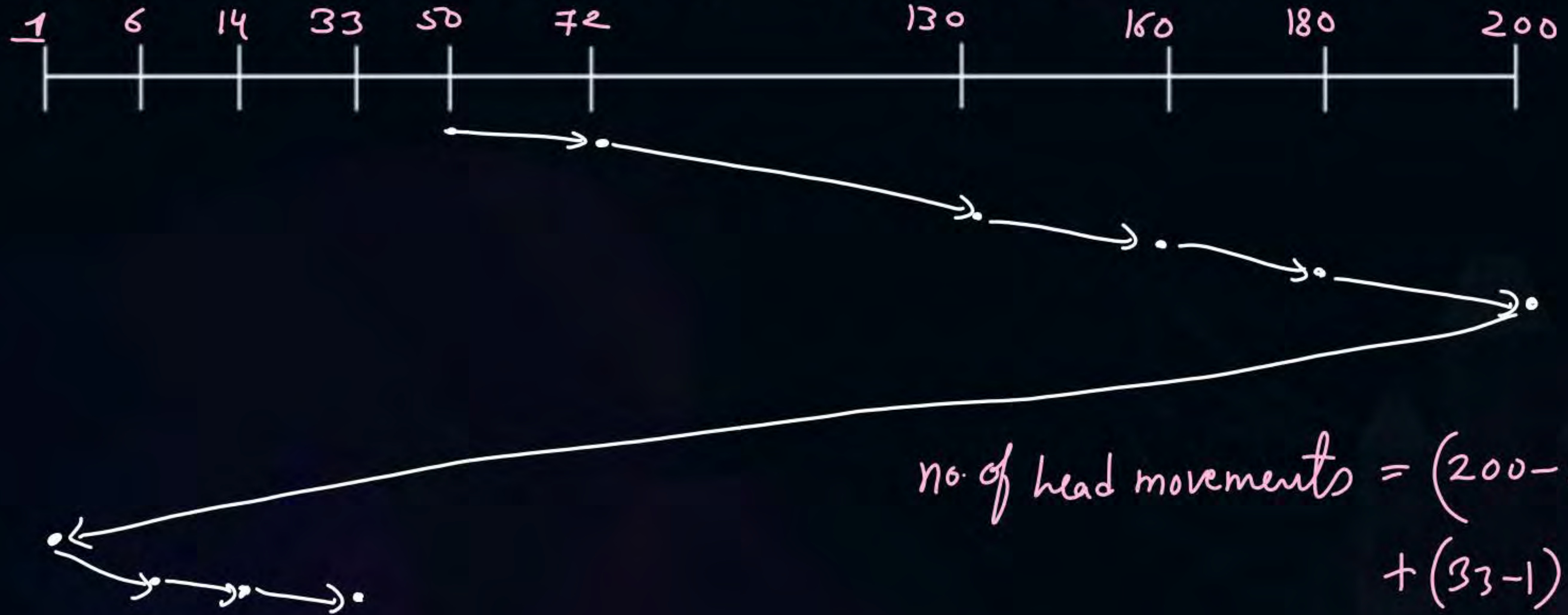
The arm should move “towards the larger value”



1 → 2 → 200

Topic : C-Scan

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



no. of head movements $= (200-50) + (200-1) + (33-1)$

$= 150 + 199 + 32$

$= 381$

Topic : C-Scan

Advantages:

- Provides more uniform wait time compared to SCAN

Topic : Look

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

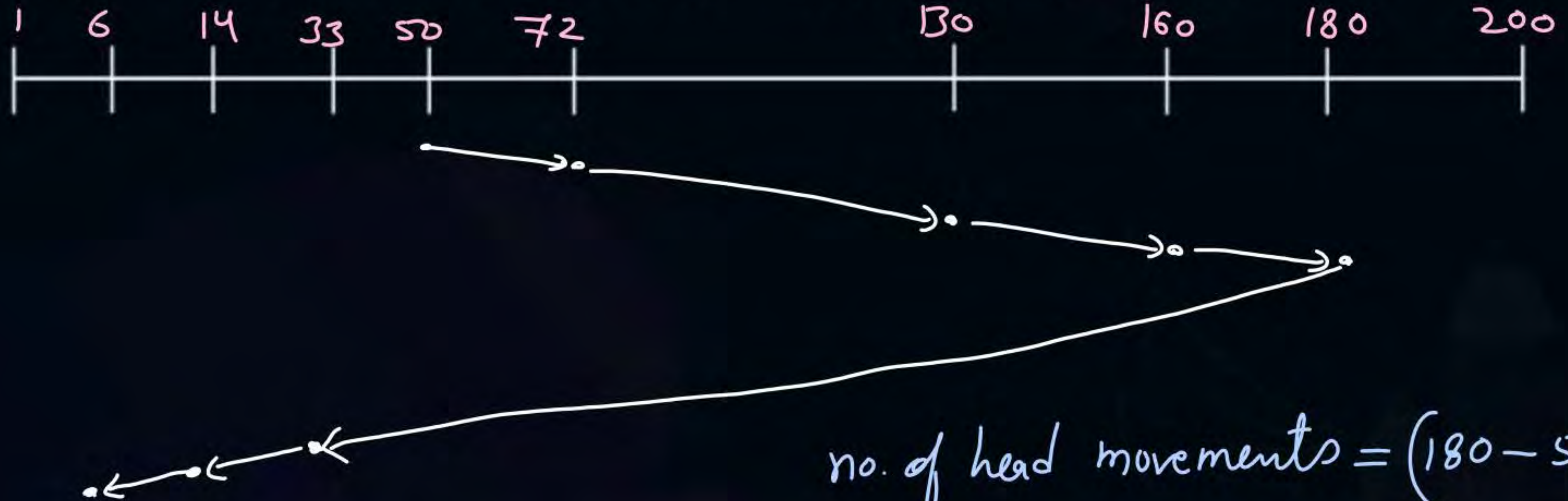
The Read/Write arm is at 50,

The arm should move "towards the larger value"

similar a scan, but the direction of head movement is changed at last cylinder no. request.

Topic : Look

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



no. of head movements $= (180-50)+(180-6)$

$= 130+174$

$= 304$

Topic : C-Look

circular-look

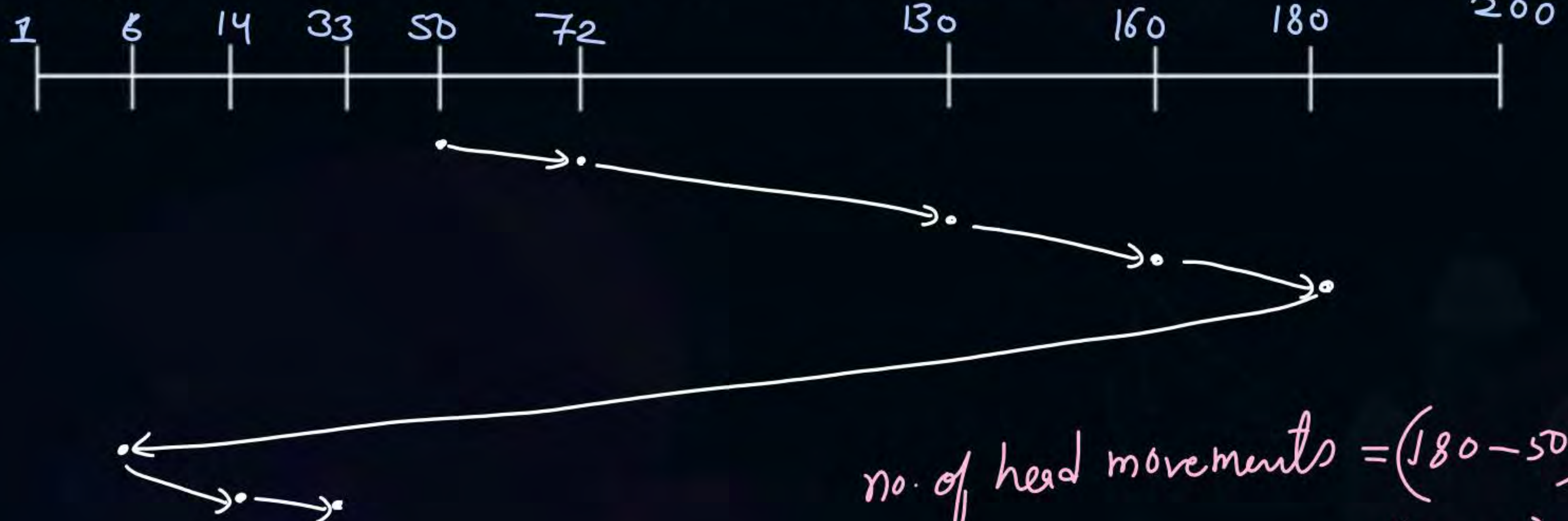
Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180

The Read/Write arm is at 50,

The arm should move "towards the larger value"

Topic : C-Look

Suppose the order of request is: 72, 160, 33, 130, 14, 6, 180



no. of head movements $= (180-50) + (180-6) + (33-6)$

$= 130 + 174 + 27$

$= 331$

ex:-

Last 2 requests are fulfilled are from cylinder no 62, 49 in the order.

49 ← 62

Head is moving towards smaller cylinder nos.

ex:- Last service at cylinder 49

and before that " 62

[MCQ]

#Q. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs? [2004]

A 50%

B 40%

C 25%

D 0%

[MCQ]

#Q. The head of a hard disk serves request following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180.

Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the request arrive before the servicing starts?

[2007]

A × 11, 139, 170, 178, 181, 184, 201, 265

B ✓ 10, 138, 170, 178, 181, 185, 201, 265

C 10, 139, 169, 178, 181, 184, 201, 265

D 10, 138, 170, 178, 181, 185, 200, 265

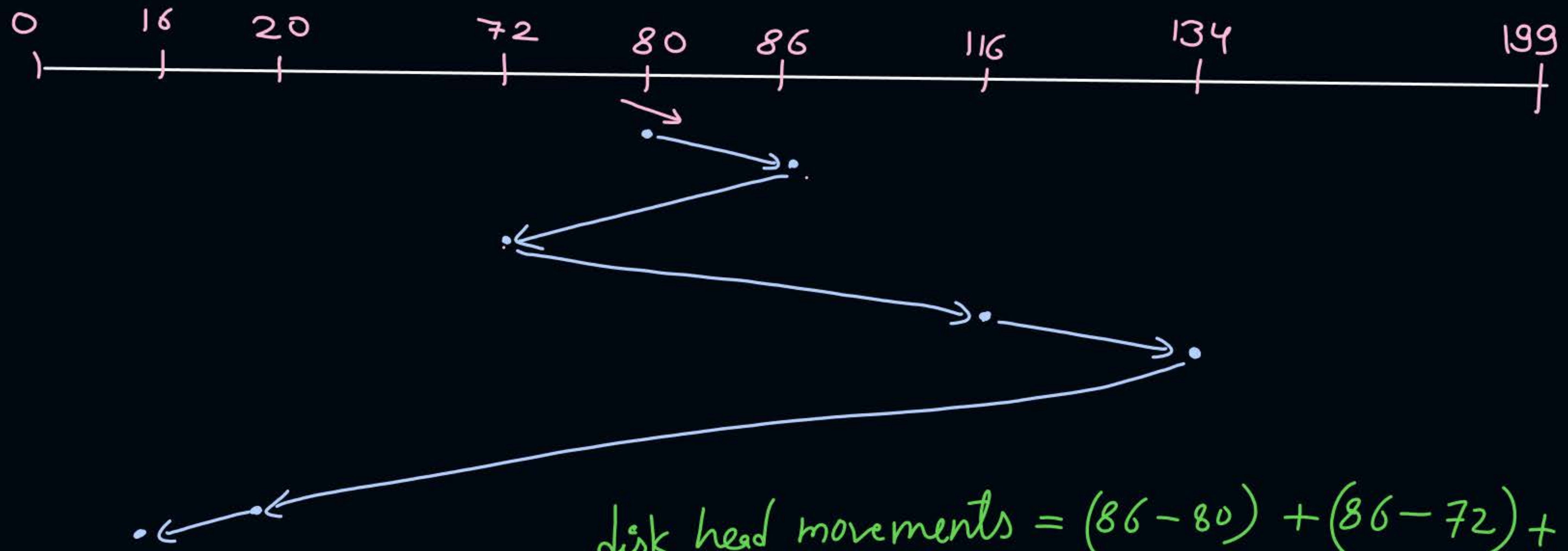
[MCQ]

Ans = 85

#Q. Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently head is positioned at sector number 100 of cylinder 80 and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____ . ? **[2018]**



$$\text{disk head movements} = (86-80) + (86-72) + (134-72) + (134-16)$$

$$= 200$$

$$\text{power consumpt}^n = \frac{20 * 200}{100} \text{ mw} + (3 * 15 \text{ mw})$$

$$= 85 \text{ mw}$$

2 mins Summary

Topic Disk Scheduling

Topic FCFS, SSTF

Topic Scan, Look Algorithms

Happy Learning

THANK - YOU